COMPLETE PROJECT DOCUMENTATION



EdAcademix

Study-Abroad Guidance + IELTS Preparation Hub
for Bangladeshi students

Choose a country • Practise IELTS • Get certified • Apply

PROJECT TYPE	Dynamic web application (PHP + MySQL + AI)
DESTINATIONS	Australia | England | Germany | Italy | USA
COVERAGE	20 universities, 3 program levels, scholarships
IELTS HUB	Listening | Reading | Writing (AI) | Speaking (live)
TEAM	Kazi Shibli Zaman Rahil | Nazifa Tabassum Hasan Al Banna | Jannatul Ferdus Setu Tahmid Ahmed Talukdar
INSTITUTION	United International University (UIU), Dhaka
DOCUMENT	Version 1.1 - 9 November 2026

Prepared from the complete project source code, database scripts and files.

Table of Contents

1. Executive Summary . . . 4

Highlights . . . 4

2. Problem Statement and Objectives . . . 5

2.1 The problem . . . 5

2.2 Objectives . . . 5

3. Evolution from GlobalEdX - What Is New . . . 6

4. Scope: What Is Built and What Is Planned . . . 6

5. Users and Access Levels . . . 7

6. Features in Detail . . . 7

6.1 Home page and navigation . . . 7

6.2 Country explorer . . . 7

6.3 University profiles (20 universities) . . . 8

6.4 Preferred University finder . . . 8

6.5 Offer Letter Request . . . 9

6.6 IELTS Prep hub . . . 9

6.7 Administration panel . . . 10

7. System Architecture and Technology . . . 11

7.1 Project structure . . . 12

8. Database Design . . . 12

8.1 Data dictionary (from the delivered database) . . . 14

8.2 Content delivered in the database . . . 15

9. Key Algorithms and Business Rules . . . 16

9.1 Next-set rotation . . . 16

9.2 Automatic marking . . . 16

9.3 AI writing evaluation . . . 16

9.4 Overall band and certificate eligibility . . . 16

9.5 Preferred University matching . . . 16

10. Security Review . . . 17

10.1 Controls already in place . . . 17

10.2 Findings and recommended fixes . . . 17

11. Installation and Run Guide . . . 18

12. Test Cases . . . 19

13. Known Limitations and Roadmap . . . 19

14. What We Built - Team and Deliverables . . . 20

15. Presentation Guide and Viva Questions . . . 21

15.1 Suggested 10-minute demo . . . 21

15.2 Likely questions and model answers . . . 21

Appendix A - Glossary **22**

1. Executive Summary

EdAcademix is an all-in-one web platform that guides Bangladeshi students from the first question - *"Where should I study abroad?"* - to the paperwork at the end: university selection, tuition and scholarship research, an **IELTS preparation hub with four practice modules**, an **AI-evaluated writing test**, a **live video speaking test with a human examiner**, a performance dashboard, an **achievement certificate**, and an **offer-letter request** that collects all application documents. Administrators run everything from a dedicated panel.

It grows out of the team's earlier GlobalEdX project (3 countries, 9 universities) and extends it to **5 countries and 20 universities** and adds the complete IELTS ecosystem (the code package is named *aspire-ielts*). It is built with PHP, MySQL, HTML/CSS/JavaScript (Bootstrap, Chart.js), FPDF for certificates, the OpenRouter/DeepSeek API for writing evaluation and Jitsi Meet for speaking sessions.

5	20	301	29	4
Countries	Universities	PHP scripts	Database tables	IELTS modules

Highlights

- **Data-driven university explorer** - for every university: fields, programs per level (Undergraduate, Graduate, PhD), tuition per level, scholarships and requirements, all from the database.
- **Preferred University finder** - choose country, program and your IELTS score to see universities whose language requirement you meet.
- **Auto-marked Reading and Listening tests** with 3 sets each, official-style band conversion and instant feedback.
- **AI examiner for Writing** - essays and letters are sent to a language model that returns a band score and written feedback.
- **Live Speaking test** - student and administrator meet in a private-named Jitsi video room; the examiner records the band score and feedback.
- **Performance dashboard** - average band, section-wise analysis, trend chart (Chart.js), dark mode and a rule-based IELTS chatbot.
- **Certificate workflow** - students with an overall band of 6 or more apply; an FPDF certificate is generated and the request goes to admins.
- **Admin panel** - users and their results, offer-letter requests with documents, certificate requests, contact messages, speaking sessions and reports.

2. Problem Statement and Objectives

2.1 The problem

Students aiming for universities abroad need two things at once: reliable information about where to apply and proof of English proficiency (IELTS). Today they use separate websites for rankings, fees, scholarships and requirements, pay for IELTS coaching without knowing their level, and fill in the same personal details and documents again and again for each application.

- Scattered, inconsistent university and fee information.
- No way to know whether the IELTS score they have meets a specific program's requirement.
- Expensive, one-size-fits-all IELTS coaching and no feedback on writing or speaking.
- Repetitive paperwork and no single place to start an application.

2.2 Objectives

#	Objective	Delivered by
1	Simplify country and university choice	5 country pages with filters (program, tuition, ranking, search) and 20 university profiles.
2	Centralise costs, scholarships and requirements	Per-university pages for tuition by level, scholarship details and admission requirements.
3	Match students to programs by IELTS score	Preferred University finder (country + program + score).
4	Provide free, structured IELTS practice	Reading, Listening, Writing, Speaking tests; video courses and tips; chatbot.
5	Give feedback beyond right / wrong	AI writing feedback and examiner speaking feedback; section analytics and trend chart.
6	Recognise achievement	Certificate application and PDF generation for students with band 6 or above.
7	Reduce application paperwork	Offer-letter request with 11 documents and an admin review screen.
8	Give staff control	Admin panel with 9 working pages.

3. Evolution from GlobalEdX - What Is New

Area	GlobalEdX (earlier)	EdAcademix (this project)
Countries	3 (Australia, Germany, USA)	5 (adds England and Italy)
Universities	9	20 (4 per country)
Program detail	Programs, fees and requirements per university	Adds separate Undergraduate / Graduate / PhD offer pages, tuition pages per level, fields of study, and scholarships
Eligibility	Recommendation Engine with weighted score	Preferred University finder using IELTS score vs language requirement
IELTS	-	Full IELTS hub: 4 test modules, video courses, tips, dashboard, chatbot, certificate
AI	Described but not built	AI writing evaluation through OpenRouter (DeepSeek)
Live interaction	-	Live speaking test (Jitsi) with administrator examiner
Administration	None (manual database)	Admin login/sign-up, user results, offers, certificates, contacts, reports, speaking
Offer letter	Form + upload	Same form with declaration, now reviewable in the admin panel
PDF	FPDF bundled, unused	FPDF used to generate certificates (TCPDF library also bundled)

4. Scope: What Is Built and What Is Planned

Area	Status	Details
Home page, country carousel, FAQ, About, Contact form	**Implemented**	index.php, about.php, contact.php (messages saved to contact_messages).
Sign up / Log in / Log out	**Implemented**	Hashed passwords, session regeneration.
Country pages x 5 with 6 filters each	**Implemented**	about_, uni_, program_, rank_, tuitionfee_, search_ for aus, eng, ger, italy, usa.
20 university profiles + contact pages	**Implemented**	40 static HTML pages in universities/.
University data pages (fields, programs, tuition, scholarships, requirements)	**Implemented**	220 PHP scripts generated from a common pattern.
Preferred University finder	**Implemented**	Preferred_university.php.
Offer Letter Request + admin review	**Implemented**	offerletter.php, submit_offer.php, Admin/admin_offer.php.
Reading and Listening tests (auto-marked)	**Implemented**	3 sets each; band conversion.
Writing test with AI scoring	**Implemented**	Needs an API key and internet access.
Speaking test (live)	**Implemented**	Needs an administrator online to join the room.
Dashboard analytics, video courses, tips, rule-based chatbot	**Implemented**	dashboard.php, Courses/, bot.php.

Area	Status	Details
Certificate application and PDF	**Implemented (needs hardening)**	cert.php, submit_cert.php, Admin/admin_cert.php (see Section 10).
GPT-powered Study-Abroad Advisor chatbot	Planned	Described in the FAQ; no code or API calls exist for it.
Automated "Smart Apply" prefill / auto-generated offer letters	Planned	Offer letters are requested and reviewed manually.
E-mail delivery of certificates	Planned	Site states 48 hours; delivery is manual (no mail code).
Online payments, counsellor chat, real-time analytics	Planned	Listed as future goals on the About page.

5. Users and Access Levels

User	Entry	Can do
Visitor	index.php	Explore countries, universities, fees, scholarships and requirements; read About and FAQ; send a contact message. Protected items show a lock.
Student	login.php / signup.php	Everything above plus the IELTS Prep dashboard, tests, courses, certificate application, Preferred University finder and Offer Letter Request.
Administrator (examiner)	Admin/admin_login.php	View all students and their test responses and results, join speaking sessions and score them, review offer-letter requests and uploaded documents, review certificate requests, read contact messages, view reports.

6. Features in Detail

6.1 Home page and navigation

The home page offers the menu **Home, Dream Countries (Australia, England, Germany, Italy, USA), IELTS Prep, Get Offer Letter, Contacts**. A carousel introduces each country. PHP reads the session: when the user is logged in, protected links are "unlocked"; otherwise a lock pop-up asks them to log in. A search box (hm_search.php) accepts a country name, verifies it in the countrylist table with a prepared statement and redirects to the matching country page (desh.html = Australia, desh2 = England, desh3 = Germany, desh4 = Italy, desh5 = USA). A shared `second_navbar.php` adds the profile menu.

6.2 Country explorer

Each country has six data pages; all are database driven and filter by country name.

Page family	Files	Shows
`about_*`	5	Country overview, currency, location, social / educational / cultural status, living style
`uni_*`	5	University cards: logo, name, location, program types, tuition category, world rank
`program_*`	5	Universities ordered by number of program levels
`tuitionfee_*`	5	Universities ordered Free, Medium, High
`rank_*`	5	Universities ordered by numeric world rank

Page family	Files	Shows
`search_*`	5	Name search with prepared statement

6.3 University profiles (20 universities)

Country	Universities
Australia	Australian National University, University of Melbourne, University of Sydney, University of Queensland
England	University of Cambridge, University of Oxford, Imperial College London, University College London
Germany	Heidelberg University, Ludwig Maximilian University of Munich, Technical University of Munich, Charité Universitätsmedizin Berlin
Italy	University of Bologna, University of Padua, Politecnico di Milano, Sapienza University of Rome
USA	Harvard University, Massachusetts Institute of Technology, Stanford University, California Institute of Technology

Every university has a description page and a contact (help-line) page in `universities/` plus a set of dynamic data pages. The per-university page families are:

Page family	Files	Content (from the database)
`fields_*`	20	Fields of study and subjects with descriptions (universityprogramfield + subject)
`pro_academic_*`	20	All subjects grouped by program level
`pro_offer_undergrate_* / graduate_* / phd_*`	60	Subjects offered at one program level (Undergraduate Programs, Graduate Programs, PhD)
`fee_academic_*`	20	Tuition summary across levels
`tuition_undergrate_* / graduate_* / phd_*`	60	Duration, number of semesters or trimesters, total tuition (USD), fee per term and fee category for one level
`scholar_academic_*`	20	Scholarship details per program and field (tuitionfee.scholarship_details)
`req_*`	20	Academic qualification, language requirement, entrance exam, other requirements, deadline

Note: the files use the spelling `undergrate`; this is only a file-name convention and is used consistently.

6.4 Preferred University finder

`Preferred_university.php` takes **country, program level and a language score**. It joins University, CountryList, admissionrequirements and Program and returns universities whose `language_requirement` either contains the entered text or starts with a number that is less than or equal to the student's score (e.g. a requirement beginning "6.5" matches a score of 7). Results are shown as cards with university, world rank, location, program and requirement.

6.5 Offer Letter Request

A multi-section form collects personal details, SSC/HSC years and medium, country / university / program / subject preference, intake season and year, and **11 documents** (NID, SSC and HSC certificates, English certificate, parents' NIDs, letter of recommendation, passport photo; passport scan, SOP and CV optional) with a final declaration and data-use consent. Files (jpg, jpeg, png, pdf) are stored in `Offer Letter/` as `Name_field.ext`; one row is inserted into offer_letter_requests. Administrators open `Admin/admin_offer.php` to list requests and view every uploaded file.

6.6 IELTS Prep hub

After login the student's **dashboard** (dashboard.php) is the hub: welcome message, "Choose a Test Section" cards (Listening, Reading, Writing, Speaking), links to Tests, Lecture Videos and Apply for Certificate, a performance section and a floating IELTS chatbot. A switch enables **dark mode**.

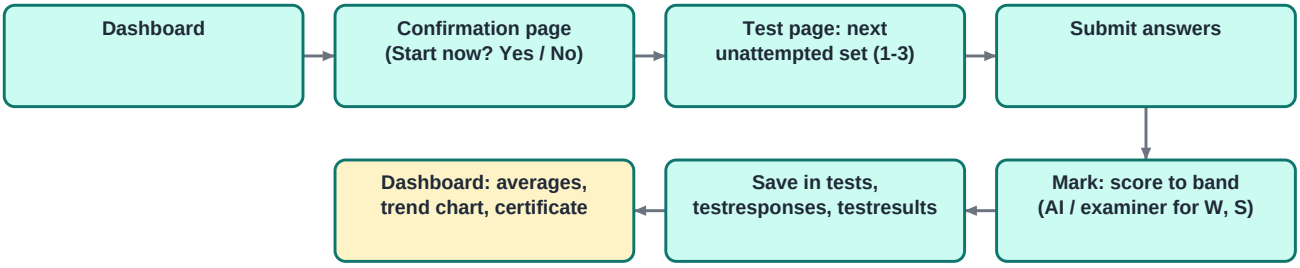


Figure 1 - One IELTS test attempt.

Module	How it works	Marking
Reading	Passages (reading_passages) and questions (questions) for the next set the student has not yet completed; set picked from 1, 2, 3.	Automatic: answers compared case-insensitively with CorrectAnswer; raw score of 40 converted to a band.
Listening	One audio file per set (Audio/listening_set1-3.mp3, path stored in listeningaudio) with an on-page timer; questions from the same table.	Automatic; separate raw-score-to-band table.
Writing	Two tasks per set (Letter Writing and Essay Writing) from writingtasks.	AI examiner: every response is sent to a language model with an IELTS-examiner prompt (Task Response, Coherence and Cohesion, Lexical Resource, Grammatical Range and Accuracy). It must return JSON {score, feedback}. The band is the average of the tasks, rounded to 1 decimal; feedback is stored per task.
Speaking	Student presses Start; a row in speaking_sessions (status ongoing) is inserted and a Jitsi Meet video room named for the student's ID opens. Admin pages list ongoing sessions and join the same room.	Human: the examiner enters a band score and feedback, saved to tests and testresults.

Band conversion (Reading - 40 questions)

Raw score	40	39	37-38	36	34-35	32-33	30-31	27-29	23-26	19-22	15-18
Band	9	8.5	8	7.5	7	6.5	6	5.5	5	4.5	4

Band conversion (Listening - 40 questions)

Raw score	39+	37+	35+	32+	30+	26+	23+	18+	16+	13+	10+
Band	9	8.5	8	7.5	7	6.5	6	5.5	5	4.5	4

Each module can be taken once per set, so a student has **three attempts per module**; after that a friendly "All Done" message is shown. Feedback texts range from "Excellent! You're close to perfect" to "Don't worry, start with easier passages".

Dashboard analytics

- Reads every row of testresults for the student: **average band** = sum of scores / number of tests.
- **Section performance** - average per Listening, Reading, Writing, Speaking.
- **Performance trend** - a Chart.js line chart of scores by test date.
- **Certificate button** becomes available when the average is 6 or higher.
- **IELTS chatbot** (bot.php) - a floating widget answering common questions (greetings, writing tips, task 1 / task 2, mistakes and more) from a predefined response dictionary; it is rule-based, not AI.

Courses and tips

For each skill there is a **Tips** page (for Reading: eight tips such as skimming first, spotting synonyms and managing time) and a **Course** page that lists video lessons in order from the sectionvideos table (section, video_number, video_path) in a no-download video player. The delivered catalogue lists **41 video lessons**: Listening 7, Reading 10, Writing 14 and Speaking 10.

Certificate

A student with an overall band of 6 or more opens `cert.php`, confirms name, e-mail, phone, date of birth and test date, and submits. The data is stored in `certificate_info`, a styled A4 certificate PDF (double border, header band, name and band) is generated with **FPDF** into `certificate/`, and the student is told the certificate will arrive by e-mail within 48 hours. Administrators review requests in `Admin/admin_cert.php`. Students below band 6 see an "Oops! Not Eligible Yet" page.

6.7 Administration panel

Page	Purpose
`admin_login / admin_signup`	Administrator authentication (hashed passwords in table admin).
`admin_dashboard`	Pick a student; see profile, tests, responses and results.
`admin_users`	Browse registered users and their activity.
`admin_speaking + check_speaking_sessions`	List ongoing speaking sessions, join the call, enter band score and feedback.
`admin_offer`	List offer-letter requests; open one to see all details and uploaded files (image preview or link).
`admin_cert`	List and view certificate applications.
`admin_contact`	Read messages from the Contact form.
`admin_report`	Per-student report with performance chart (score by date).

7. System Architecture and Technology

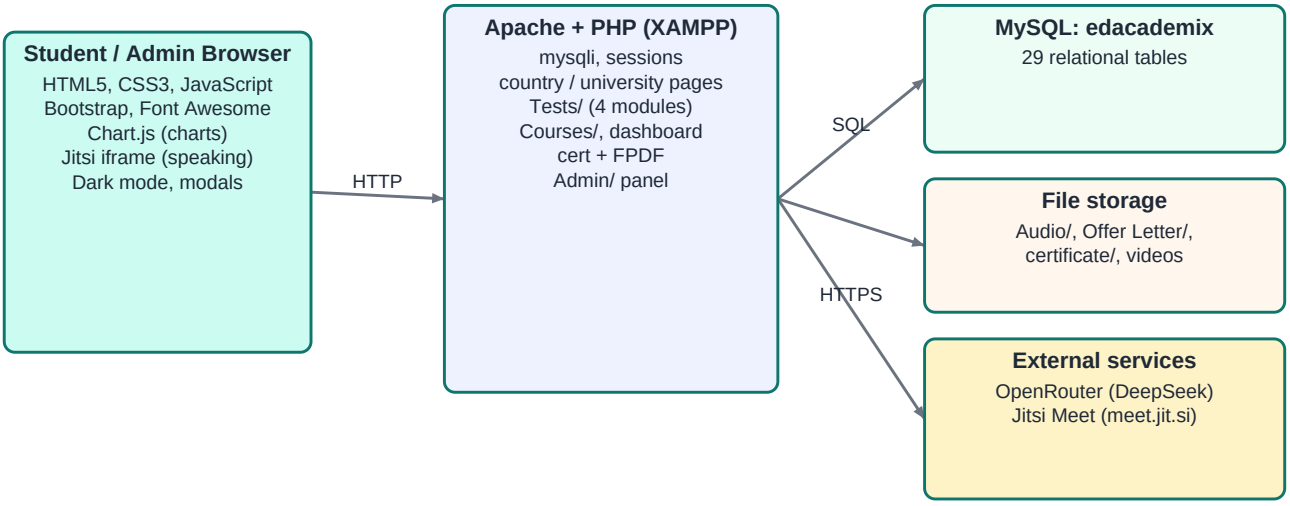


Figure 2 - Architecture of EdAcademix.

Layer	Technology	Purpose
Front end	HTML5, CSS3, JavaScript, Bootstrap 5, Font Awesome, Chart.js, Google Fonts	Responsive pages, charts, modals, dark mode
Back end	PHP (mysqli, sessions, cURL)	Business logic, marking, API calls
Database	MariaDB 10.4 / MySQL, database **edacademix**	29 tables
PDF	FPDF (used), TCPDF (bundled)	Certificate generation
AI	OpenRouter chat-completions API, DeepSeek chat model	Writing evaluation returning JSON
Video	Jitsi Meet external API	Live speaking room
Packages	Composer: symfony/http-client, openai-php/client, vlucas/phpdotenv, nyholm/psr7	Declared in composer.json; the code calls the API directly with cURL, so these are not yet used
Server and tools	XAMPP, phpMyAdmin, Visual Studio Code	Development environment

7.1 Project structure

Path	Contents
`/ (301 PHP scripts in total)`	Core pages (index, login, signup, dashboard, tests hub, offer letter, certificate, contact, about, videos, bot) and the 250+ data pages
`Tests/`	12 scripts: reading, listening, writing, speaking tests, confirmations and submit handlers
`Courses/`	8 scripts: tips and course playlists per skill
`Admin/`	10 scripts: admin panel
`universities/`	40 HTML pages (20 universities + 20 contact pages), css, js, images
`Audio/`	3 listening recordings (about 119 MB)
`Videos/`	41 MP4 lessons (listening 7, reading 10, writing 14, speaking 10). Large; not stored in Git - see Videos/README.md
`database/, docs/`	edacademix.sql (sanitized), logo and this documentation
`certificate/ , Offer Letter/`	Generated certificates and uploaded documents (empty folders; contents are git-ignored)
`Tests/config.example.php`	Template for the API key file config.local.php (git-ignored)
`css/, js/, img/, images/, webfonts/`	Styling, scripts, logos and photos
`fpdf/, tcpdf/, composer.json`	Third-party libraries and Composer manifest (vendor/ is not stored)

8. Database Design

The database **edacademix** (MariaDB 10.4 / MySQL, utf8mb4, InnoDB) has **29 tables**. The file `database/edacademix.sql` contains the complete structure and all catalogue and IELTS test content. To protect privacy, every row of user, administrator, certificate, offer-letter, contact, test-result and answer data was **removed** before publishing; those tables are present but empty.

Group	Tables
Study abroad (9)	countrylist, countrydetails, university, program, fieldofstudy, subject, universityprogramfield, tuitionfee (with scholarship_details), admissionrequirements
IELTS content (7)	questions, reading_passages, listeningaudio, writingtasks, sectionvideos, speakingprompts*, bandscorecriteria*
IELTS attempts and sessions (8)	tests, testresults, testresponses, writingresponses, speaking_sessions, listeningresponses*, speakingrecords*, userprogress*
Accounts and operations (5)	users, admin, offer_letter_requests, certificate_info, contact_messages

* These five tables exist in the schema but are **not used by the current code** (reserved for future features).

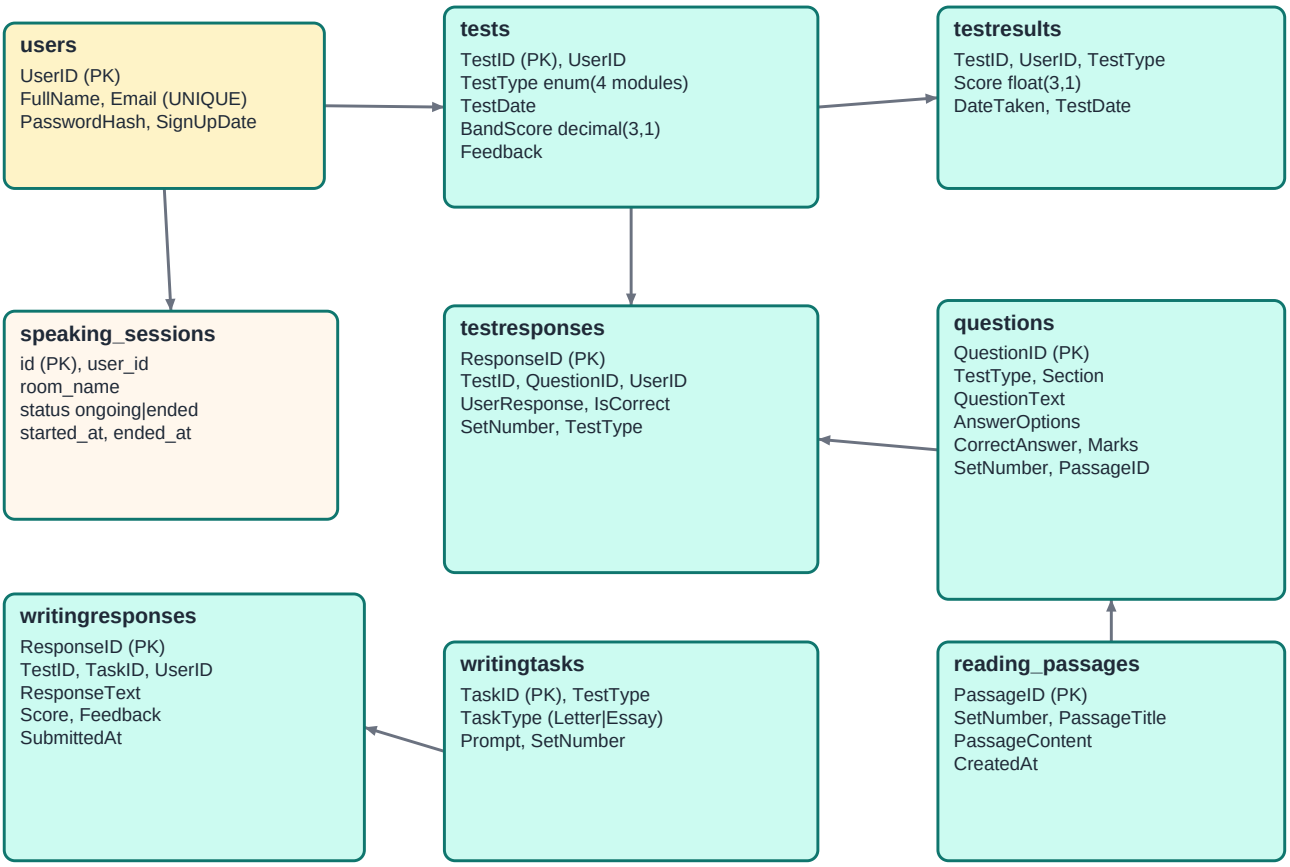


Figure 3 - IELTS part of the schema (real structure). listeningaudio (Section, AudioFilePath, Transcript, SetNumber) and sectionvideos (section, video_number, video_path) are standalone lookup tables.

8.1 Data dictionary (from the delivered database)

Table	Columns	Purpose	Rows
admin	AdminID, Username, PasswordHash, email	Administrator accounts	empty
admissionrequirements	requirement_id, university_id, program_id, field_id, academic_qualification, language_requirement, entrance_exam, other_requirements, application_deadline	Entry requirements per university, program level and field	156
bandscorecriteria	CriteriaID, TestType, CriteriaDescription, MaxScore	Reserved - not used	empty
certificate_info	id, user_id, full_name, email, phone_number, date_of_birth, overall_band_score, test_date, submitted_at	Certificate applications	empty
contact_messages	id, name, email, message, submitted_at	Contact-form messages	empty
countrydetails	details_id, country_id, overview, currency, location, social_status, educational_status, cultural_status, living_style	About-page text per country	5
countrylist	country_id, name	The 5 countries	5
fieldofstudy	field_id, name	Fields of study	3
listeningaudio	AudioID, Section, AudioFilePath, Transcript, SetNumber	One audio file per listening set	3
listeningresponses	id, user_id, QuestionID, Answer, test_set, submitted_at	Reserved - not used	empty
offer_letter_requests	id, full_name, father_name, mother_name ... already_applied (33 columns)	Offer letter requests (personal data and 11 file paths)	empty
program	program_id, name	Program levels (Undergraduate, Graduate, PhD)	3
questions	QuestionID, TestType, Section, QuestionText, AnswerOptions, CorrectAnswer, Marks, SetNumber, PassageID	Reading and Listening questions with options and correct answer	240
reading_passages	PassageID, SetNumber, PassageTitle, PassageContent, CreatedAt	Reading passages (3 per set)	9
sectionvideos	id, section, video_number, video_path	Course video playlist	41
speaking_sessions	id, user_id, room_name, started_at, ended_at, status	Live speaking sessions	empty
speakingprompts	PromptID, Part, PromptText	Reserved - not used	empty
speakingrecords	RecordID, TestID, FilePath	Reserved - not used	empty
subject	subject_id, name, field_id, description	Subjects with descriptions	184
testresponses	ResponseID, TestID, QuestionID, UserResponse, IsCorrect, SetNumber, TestType, UserID	Every answer given in Reading / Listening	empty

Table	Columns	Purpose	Rows
testresults	TestID, UserID, TestType, Score, DateTaken, TestDate	Final score per attempt	empty
tests	TestID, UserID, TestType, TestDate, BandScore, Feedback	Attempt header with band and feedback	empty
tuitionfee	fee_id, university_id, program_id, field_id, duration, tuition_fee, semesters_or_trimesters_fee, scholarship_details, semesters_or_trimesters, fee_category	Duration, tuition, fee per term, category, scholarship details	156
university	university_id, name, country_id, location, global_rank	The 20 universities	20
universityprogramfield	upf_id, university_id, program_id, field_id, subject_id	Which university offers which subject at which level	360
userprogress	ProgressID, UserID, Module, AttemptCount, AverageScore, LastTestDate, ImprovementSuggestions	Reserved - not used	empty
users	UserID, FullName, Email, PasswordHash, SignUpDate	Student accounts	empty
writingresponses	ResponseID, TestID, TaskID, UserID, ResponseText, Score, Feedback, SubmittedAt	Writing answers with AI score and feedback	empty
writingtasks	TaskID, TestType, TaskType, Prompt, SetNumber	Letter and essay prompts per set	6

8.2 Content delivered in the database

Content	Amount
Countries / universities	5 / 20
Subjects / program-field links	184 / 360
Tuition-and-scholarship rows / admission-requirement rows	156 / 156
Reading and Listening questions	240 (120 each: 3 sets of 40)
Reading passages / writing tasks / listening audio entries	9 / 6 / 3
Video lesson entries (files are not stored in Git)	41 (listening 7, reading 10, writing 14, speaking 10)

9. Key Algorithms and Business Rules

9.1 Next-set rotation

For each module the system reads the distinct `SetNumber` values the student already has in testresponses (or writingresponses joined to writingtasks) and offers the first of 1, 2, 3 that is missing. If all three are present the student sees the "All Done" message. This gives a fixed, repeatable practice path without extra state.

```
for ($i = 1; $i <= 3; $i++) {
    if (!in_array($i, $completed_sets)) { $current_set = $i; break; }
}
```

9.2 Automatic marking

Correct answers are loaded server-side (`SELECT QuestionID, CorrectAnswer FROM questions`). Each submitted answer is trimmed and lower-cased and compared; a correct answer adds one to the raw score and every response is stored with `IsCorrect`. The raw score is mapped to a band by the tables in Section 6.6, a row is added to `tests` (header), `testresponses` (each answer) and `testresults` (final score), and a feedback sentence is chosen from the band range.

9.3 AI writing evaluation

1. Load the two writing tasks (Letter Writing, Essay Writing) of the next unattempted set.
2. For each task build a chat request: a *system* message describing the IELTS examiner's four criteria and demanding `{"score": number, "feedback": string}` only; a *user* message with the prompt and the student's text.
3. Send it with cURL to the OpenRouter chat-completions endpoint, read `choices[0].message.content` and parse the JSON (with a second attempt that strips quotes if the first parse fails).
4. Save score and feedback in writingresponses; average the task scores to one decimal for the final band; save it in testresults and tests.

9.4 Overall band and certificate eligibility

Overall band = average of all stored scores. Example: Listening 6.5, Reading 7.0, Writing 6.0, Speaking 6.5 gives (6.5 + 7.0 + 6.0 + 6.5) / 4 = **6.50**; because it is at least 6 the certificate button is enabled.

9.5 Preferred University matching

```
WHERE LOWER(c.name)=LOWER(?) AND LOWER(p.name)=LOWER(?)
  AND ( LOWER(ar.language_requirement) LIKE CONCAT('%', LOWER(?), '%')
        OR CAST(SUBSTRING_INDEX(LOWER(ar.language_requirement),' ',1) AS DECIMAL) <= ? )
```

The first word of the requirement text (for example "6.5" in "6.5 IELTS overall") is cast to a number and compared with the student's score.

10. Security Review

10.1 Controls already in place

- Passwords for students and admins are hashed with `password_hash` and verified with `password_verify`; the student login regenerates the session ID.
- Prepared statements are used for authentication, test marking, searches, contact messages and offer-letter lookups; admin IDs taken from the URL are cast to integers.
- Output on listing pages uses `htmlspecialchars()`; the contact form validates the e-mail.
- Admin pages check `$_SESSION['AdminID']` and redirect to the admin login when missing.
- Upload extension allow-list and video player with downloads disabled.

10.2 Findings and recommended fixes

The review found the following issues. Items marked Critical or High should be fixed before any public use.

Priority	Where	Finding	Recommended fix
Critical (resolved in this repository)	Tests/submit_writing.php, Tests/.env	Earlier versions kept an API key as a fallback value in the source and in a .env file inside the web root (the .env was never loaded). In this repository the key is removed: the code reads DEEPSEEK_API_KEY from the environment or from Tests/config.local.php, which is git-ignored.	**Revoke the old key** at OpenRouter and use a new one; never commit config.local.php.
High	admin_signup.php, Admin/admin_signup.php	Anyone can open the page and create an administrator account.	Remove public admin sign-up; create admins by script or only from a logged-in super-admin.
High	cert.php	Eligibility trusts the `?score=` URL parameter, so any user can type score=9. The redirect target certificate_restriction.php does not exist (file is cert_restrict.php).	Compute the average on the server from testresults for the logged-in user; fix the redirect.
High	Admin/admin_speaking.php, check_speaking_sessions.php	No admin-session check; anyone can list live sessions or POST a band score for any user ID.	Add the AdminID guard and CSRF token; validate score range 0-9.
High	Tests/*.php, bot.php	Test pages read $_SESSION['UserID'] without checking it; pages redirect to login_form.html, which does not exist.	Shared auth include that redirects to login.php.
High	submit_offer.php	Documents stored in a web-reachable folder with predictable names and only extension checks; folder created with 0777.	Store outside the web root with random names, MIME and size checks, authorised download.
Medium	Reading / Listening submit	The band is bound as an integer (`"iisiss"`) when saving to testresults.Score (float(3,1)), so 6.5 is stored as 6 and dashboard averages come out slightly low. tests.BandScore is updated correctly with d.	Bind the band with d (use "iisdss").

Priority	Where	Finding	Recommended fix
Medium	speaking_sessions / speaking_test.php	The query uses ON DUPLICATE KEY UPDATE, but the table has only a primary key on id (no UNIQUE key on user_id), so every start inserts a new row instead of refreshing one.	Add UNIQUE(user_id) or update by user_id.
Medium	Offer Letter/	Uploaded documents are web-reachable (the admin page links to them directly) and are stored with predictable names.	Store outside the web root and serve through an authorised download script.
Medium	Speaking	Room names are predictable (EdAcademixIELTS_SpeakingTest_User + ID) on the public Jitsi server.	Random room tokens or Jitsi JWT authentication.
Medium	Many files	Database credentials (root, no password) repeated; connect.php host string has a stray space; some pages open their own connection.	One config outside the web root; restricted MySQL user.
Medium	signup.php	Minimum password length is 3.	Enforce 8+ characters.
Low	index.php, profile.php	Leftover placeholder session values (a fixed name and e-mail) and the e-mail is never stored at login.	Store name and e-mail from the database at login.
Low	Preferred_university .php	Database values echoed without escaping.	Escape output.
Low	composer.json	OpenAI and HTTP packages are declared but not used; no vendor folder.	Use phpdotenv / HTTP client or remove the dependencies.

11. Installation and Run Guide

1. Install XAMPP (PHP 8, Apache, MySQL). In php.ini enable `curl` and make sure outgoing HTTPS works (needed for AI writing evaluation).
2. Clone or copy the project to `C:\xampp\htdocs\EdAcademix`.
3. Start Apache and MySQL, open phpMyAdmin, create database `edacademix` (utf8mb4_general_ci) and **import** `database/edacademix.sql`. It contains all 29 tables plus the catalogue and IELTS test content.
4. Check `connect.php` (localhost, root, empty password, edacademix) and the repeated settings in Courses/ and contact.php.
5. Copy your `Videos` folder (41 MP4 files named listening_1 ... writing_14) into the project folder next to `Audio`. The 3 listening MP3 files are already in `Audio/`.
6. For AI writing evaluation: copy `Tests/config.example.php` to `Tests/config.local.php` and paste your own OpenRouter API key (or set the environment variable `DEEPSEEK_API_KEY`).
7. Make sure `Offer Letter/` and `certificate/` are writable by Apache.
8. Open `http://localhost/EdAcademix/index.php`. Create the first administrator at `/admin_signup.php`, then remove or protect that page.

12. Test Cases

ID	Test	Steps	Expected result
T1	Sign up / log in	Register, log in	Redirect to index; name in profile menu
T2	Locked features	As visitor click IELTS Prep	Access Required pop-up
T3	Country search	Type Italy in the home search	desh4.html opens
T4	Tuition order	Open tuitionfee page	Free, Medium, High
T5	Level pages	Open MIT, Graduate tuition and offers	Only Graduate rows
T6	Preferred finder	Germany, Graduate Programs, 7.0	Universities whose requirement is 7 or lower
T7	Reading set rotation	Submit reading three times	Sets 1, 2, 3 then All Done
T8	Band conversion	Reading raw 34	Band 7
T9	Listening audio	Open listening test	Set audio plays; timer visible
T10	AI writing	Submit two answers (key configured)	Two feedback blocks and averaged band
T11	Speaking	Student starts; admin opens ongoing sessions	Session listed; both join; band saved by admin
T12	Dashboard	After tests open dashboard	Average, section averages and trend chart
T13	Certificate	Average below 6 / 6 or above	Not-eligible page / form and PDF in certificate/
T14	Offer letter	Submit with all required files	Row saved; admin can open files
T15	Admin guard	Open admin_dashboard without login	Redirect to admin login

13. Known Limitations and Roadmap

- Five schema tables (listeningresponses, speakingprompts, speakingrecords, bandscorecriteria, userprogress) are not used by the current code.
- The 41 course videos and the 3 listening recordings are large files; the videos are not stored in Git.
- Each test set can be taken only once (3 attempts per module).
- AI scoring depends on a third-party model and internet; scores are indicative, not official IELTS results.
- Speaking needs an examiner online at the same time.
- Hundreds of near-identical data pages (250+); one parameterised template would remove most of them.
- FAQ promises (GPT advisor, Smart Apply, e-mailed certificates) are not implemented.
- Overall band mixes all attempts, not the best or latest per module.

Phase	Improvement	Benefit
1	Fix the High items in Section 10, add UNIQUE(user_id) to speaking_sessions, correct the band binding, central configuration	Safe, consistent base
2	Real certificate verification (server-side average, unique ID / QR) and e-mail delivery (PHPMailer)	Trustworthy certificates

Phase	Improvement	Benefit
3	Speaking booking calendar, recordings and rubric forms; random Jitsi rooms	Scalable speaking tests
4	More sets, timed sections, question types, adaptive practice	Better practice
5	Real study-abroad advisor chatbot grounded in the platform's database	Delivers the stated AI vision
6	Admin CRUD for universities, fees and scholarships; one template for all data pages	No manual DB edits
7	More countries, mobile app, payments, counsellor chat	Growth

14. What We Built - Team and Deliverables

EdAcademix was created by a team of five at **United International University (UIU), Dhaka**. The names and contacts below are those listed on the website's Contact page.

Team member	Contact (from the website)	Responsibility
Kazi Shibli Zaman Rahil	iamksrahil03@gmail.com	[add your role]
Nazifa Tabassum	rawshanniva14@gmail.com	[add your role]
Hasan Al Banna	hasanalbanna01234@gmail.com	[add your role]
Jannatul Ferdus Setu	jannatulsetu68@gmail.com	[add your role]
Tahmid Ahmed Talukdar	tahmidahmed789456@gmail.com	[add your role, e.g. IELTS hub / AI integration]

Deliverable	Detail
Application code	301 PHP scripts (about 240,000 lines including inline HTML / CSS): 271 root, 12 tests, 8 courses, 10 admin
Static pages	5 country pages and 40 university pages
Database	29 tables; delivered content: 5 countries, 20 universities, 156 fee rows, 156 requirement rows, 184 subjects, 240 IELTS questions, 9 passages, 6 writing tasks, 41 video entries
IELTS hub	4 test modules, 3 sets each, 3 listening recordings, video courses, tips, dashboard, chatbot
Integrations	OpenRouter / DeepSeek for writing, Jitsi Meet for speaking, FPDF for certificates
Admin panel	9 pages for users, results, speaking, offers, certificates, contacts and reports
Media	Logos, country and university images, 3 listening recordings (119 MB) and 41 video lessons

15. Presentation Guide and Viva Questions

15.1 Suggested 10-minute demo

1. **Problem (1 min)** - scattered information plus separate IELTS preparation.
2. **Explorer (2 min)** - choose Germany, filter by fees, open TUM: fields, Graduate tuition, scholarships, requirements.
3. **Preferred finder (1 min)** - enter an IELTS score and show matches.
4. **IELTS hub (3 min)** - take a short Reading set, show the band; submit a Writing answer and show AI feedback.
5. **Speaking (1 min)** - start a session, show the admin joining.
6. **Dashboard and certificate (1 min)** - average, trend chart, certificate PDF.
7. **Admin and roadmap (1 min)** - offer-letter review; future work.

15.2 Likely questions and model answers

Q. How is EdAcademix different from GlobalEdX?

It extends GlobalEdX from 3 to 5 countries and 9 to 20 universities and adds scholarships, per-level pages, the complete IELTS hub, AI writing feedback, live speaking, certificates and an admin panel.

Q. How do you grade Reading and Listening?

Answers are compared case-insensitively with CorrectAnswer from the database; the raw score out of 40 is converted to a band using threshold tables, and every answer is stored with IsCorrect.

Q. How does the AI writing evaluation work?

The student's text and the task prompt are sent to a language model with a system prompt that acts as an IELTS examiner and must reply in JSON {score, feedback}. We parse it, store it and average the two tasks.

Q. Is the AI score official?

No. It is a practice estimate; official IELTS scores come only from the test centre.

Q. How does the speaking test work?

Starting the test creates a speaking_sessions row and opens a Jitsi Meet room named after the student; the administrator lists ongoing sessions, joins the same room, and records band and feedback.

Q. How are test sets chosen?

The system finds the first set number (1-3) not yet present in the student's responses, so each student works through the sets in order.

Q. Why can the certificate be misused today?

cert.php reads the score from the URL. We documented the fix: compute the average on the server from testresults.

Q. How is the dashboard calculated?

Average = sum of all scores / count; section averages group by test type; the Chart.js line chart plots score by test date.

Q. How do you keep API keys safe?

The key is read from an environment variable or from Tests/config.local.php, which is git-ignored. An earlier version had a fallback key in the source; it was removed and the old key must be revoked.

Q. Why is the speaking session table not unique per user?

The code uses ON DUPLICATE KEY UPDATE, but the table only has a primary key on id, so each start adds a row. Adding UNIQUE(user_id) fixes it; we list it as a known issue.

Q. How does the Preferred University finder compare IELTS?

It casts the first word of language_requirement to a decimal and compares it with the entered score, or matches the text.

Q. Why so many PHP files?

Each university has separate pages per level generated from one pattern. A single parameterised template would be the next refactor.

Q. What are the limitations?

Single-use sets, dependence on an external AI service, unused reserved tables, and the security items in Section 10.

Q. How would you scale it?

Move to a framework with a central data layer, add caching, a queue for AI calls, WebRTC rooms with authentication, and a mobile app.

Appendix A - Glossary

Term	Meaning
IELTS	International English Language Testing System; bands 0-9 for Listening, Reading, Writing, Speaking
Band score	IELTS result scale in half-point steps
Raw score	Number of correct answers out of 40 before conversion to a band
Jitsi Meet	Open-source video conferencing service used for the speaking room
OpenRouter / DeepSeek	API gateway and language model used to evaluate writing
FPDF	PHP library that creates PDF files (certificates)
Prepared statement	SQL sent separately from its data to prevent injection
SSC / HSC	Bangladesh secondary and higher-secondary certificates